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TiF: A Multi-Scale Data Fusion and Fourier Encoding Framework for Financial Risk Prediction

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ABSTRACT

Financial forecasting often treats annual, quarterly, and monthly information in isolation, which harms robustness in volatile markets. We present *Time-integrated Fourier (TiF)*, a lightweight framework that fuses annual, quarterly, and monthly signals via energy-based Fourier encoding and a compact 1D-CNN backbone with learnable cross-timescale gating. TiF is evaluated under a standardized rolling-origin expanding-window protocol on a core technology benchmark (Apple, Microsoft, Adobe) and further examined on two external validations: (A) a non-financial EBITDA cohort ($N = 105$) and (B) a finance-sector cohort with ROAA as the target. Across studies, some baselines attain the best single-metric scores (e.g., a Transformer for R^2, CatBoost for RMSE), whereas TiF delivers consistently strong and *balanced* performance over R^2/RMSE/MAPE with small variance and modest parameter counts. Ablations quantify the contribution of Fourier bands and each timescale branch, and robustness checks confirm the same qualitative ranking across winsorization/clipping settings. All experiments share identical preprocessing, validation length, and hyperparameter budgets, and we report mean $\pm$ SD across rolling origins and seeds to ensure fair comparison.

1 | Introduction

In modern corporate management, financial decision-making faces an increasingly complex and highly dynamic environment. The acceleration of globalization, technological advancement, and the rise of the digital economy have amplified the multidimensional risks that firms must navigate (Brown and Rocha 2020; Kwon 2022; Jiang et al. 2023; Brynjolfsson and McAfee 2017). Managers are thus challenged to balance short-term financial performance with long-term strategic objectives (Rappaport 1986; Graham et al. 2005; Laverty 1996; Wu et al. 2021).

1.1 | Limitations of Traditional Approaches

Conventional financial analysis relies on statistical models such as linear regression (Fama and French 1992; Stock and Watson 2007), ARIMA (Box et al. 1976), and GARCH (Bollerslev 1986). While effective for specific tasks, these models often assume single-frequency data (typically annual or quarterly) and struggle with nonlinear dynamics, heteroskedasticity, and the integration of multi-timescale information (Graham et al. 2005; Kristjanpoller and Minutolo 2015; Hyndman and Athanasopoulos 2018). Such limitations become especially acute in volatile, data-rich environments where financial instability can be triggered by overlooked short-term signals (Laverty 1996; Stein 1989; Hendricks and Singhal 2005; Longstaff 2010).

1.2 | Recent Advances in Machine Learning

The emergence of machine learning has brought new tools to financial risk forecasting. Random forests, SVMs, and more recently deep learning architectures—such as CNNs, LSTMs, and Transformers—demonstrate strong capacity for extracting nonlinear patterns from complex, high-dimensional data (Breiman 2001; Sezer and Ozbayoglu 2018; Borovykh et al. 2017; Fischer and Krauss 2018; Vaswani et al. 2017; Lim et al. 2021; Chen and Guestrin 2016; Ke et al. 2017; Zhou 2012). However, most studies continue to focus on feature extraction from a single time scale, leaving the integration of monthly, quarterly, and annual information underexplored (Zhang et al. 2022; Lee 2007). Although multimodal fusion techniques such as copula models and transformer-based frameworks are emerging (Tang and Matteson 2021; Zhou et al. 2021), the simultaneous handling of high-dimensional features and multi-timescale temporal coupling remains a core research challenge (Cao et al. 2024; Kristjanpoller and Minutolo 2015).

1.3 | The Promise of Multi-Timescale and Frequency-Domain Modeling

Recent work demonstrates that financial data across different temporal resolutions (daily, monthly, quarterly, annual) convey complementary information—short-term data reflect real-time market fluctuations, while long-term records reveal strategic stability (Laverty 1996; Wu and Li 2006). Multi-scale deep learning architectures, such as MS-CNNs and transformers, and frequency-domain techniques (e.g., Fourier encoding, variational mode decomposition) have shown promise in capturing both local and global patterns, uncovering nonlinear interactions, and improving forecasting accuracy (Zhou et al. 2021; Cao et al. 2024; Zhang et al. 2022; Mallat 1999). Yet, most existing frameworks are limited in their ability to jointly model multiple temporal scales and integrate heterogeneous features in a unified system.

1.4 | Our Motivation and Research Questions

To address these challenges, this study investigates the following questions:

1. How can multi-timescale financial data (annual, quarterly, monthly) be effectively integrated for robust risk analysis and forecasting (Zhang et al. 2022)?
2. How can advanced feature encoding techniques—such as Fourier encoding—be used to capture nonlinear and cross-scale dependencies (Cao et al. 2024)?
3. How can deep learning and data fusion strategies enhance the robustness and accuracy of financial risk prediction (Zhou et al. 2021)?

1.5 | Summary of Contributions

In response, we propose the *Time-integrated Fourier (TiF)* model, a novel financial risk prediction framework that unifies multi-timescale data fusion, Fourier-based feature encoding, and deep CNN architectures. The main contributions are as follows:

- We introduce a unified modeling framework that integrates annual, quarterly, and monthly data using hierarchical fusion and frequency-domain representation.
- We combine Fourier encoding with deep CNNs to accurately capture nonlinear, cross-scale coupling in financial data.
- We develop a multimodal fusion strategy that incorporates operational, macroeconomic, and sentiment indicators, enhancing prediction robustness and generalizability.
- We conduct comprehensive experiments—performance evaluation, ablation, generalization, and baseline comparisons—to demonstrate the practical effectiveness of TiF in financial risk forecasting.

This work provides methodological advances and practical guidance for the integration of multi-scale, high-dimensional financial data, offering a powerful tool for corporate decision-makers navigating an increasingly volatile market environment.

2 | Data and Targets

2.1 | Sources and Coverage

We use *Compustat* fundamentals (annual `funda`, quarterly `fundq`) and security—monthly records (`secm`) via WRDS/S&P Global.[1] Fiscal calendars are respected: Quarterly/annual items align to fiscal period ends; monthly series are synchronized by fiscal month within each firm. The core benchmark covers Apple, Microsoft, and Adobe; two external cohorts test transferability.

2.2 | Targets

We study EBITDA margin and ROAA:

$$\text{EBITDA margin}_t = \frac{\text{EBITDA}_t}{\text{Revenue}_t}, \tag{1}$$

$$\text{ROAA}_t = \frac{\text{Net Income}_t}{\text{Average Total Assets}_t}. \tag{2}$$

Annual targets are evaluated; quarterly/monthly signals serve as inputs. Targets undergo $(1\%, 99\%)$ winsorization and hard clipping: $[-2, 2]$ for EBITDA margin and $[-0.2, 0.2]$ for ROAA. For each firm, we record the fraction of clipped values (`clip_frac`) as a stability indicator.

2.3 | Cohort Construction

Exclusions: (i) Funds/ETFs and similar entities via name heuristics; (ii) firms with missing identifiers or inconsistent fiscal calendars. **Length/variance checks:** a minimum history of valid annual targets (`min_years` $\in \{10, 12\}$) and a non-degenerate validation segment. With v the validation length, firms with validation-window standard deviation `val_std` $< 10^{-4}$ are removed. **Robust target screening:** after winsorization/clipping, drop firms with `clip_frac` > 0.5.

2.3.1 | Cohorts Used

- **Core Technology Benchmark** (AAPL, MSFT, ADBE): used for detailed, per-model comparisons under a unified protocol.
- **External I:** Non-financial cohort (EBITDA): Representative company-level summary ($N = 3$).

- **External II: Financials (ROAA).** Finance-sector firms (SIC 6000–6999) with the same screening pipeline. Because ROAA series undergo stronger clipping/variance checks, N is smaller than in the non-financial cohort.

2.4 | Feature Families (Overview)

All features use information available at (or before) each origin and are standardized with training-window statistics only (Section 3). Branches include profitability, leverage, cash-flow, growth, and return/volatility indicators derived from Compustat items; exact lists and formulae will be released. Hyperparameter budgets are matched across models for fairness.

2.4.1 | Reproducibility Artifacts

We release cohort CSVs, `target_diagnose.csv`, configuration files (winsor/clipping bounds, `min_years`), and scripts to rebuild cohorts from raw WRDS pulls.

3 | Methods

We propose *Time-integrated Fourier* (TiF), which (i) encodes each time-scale branch (annual, quarterly, monthly) in the frequency domain with energy-based truncation and light smoothing; (ii) maps branch embeddings with a lightweight 1D-CNN; and (iii) fuses branches via a learnable cross-timescale gate. All preprocessing is re-fit at every rolling origin to avoid look-ahead leakage.

3.1 | Notation and Preprocessing

Let $b \in \{\text{year}, \text{quarter}, \text{month}\}$ index a time-scale branch. For a firm, let $X^{(b)} \in \mathbb{R}^{T_b \times d_b}$ be time-ordered features and $y \in \mathbb{R}^{T_y}$ the annual target. Standardization uses *training-window* statistics only:

$$\tilde{X}^{(b)}_{t,j} = \frac{X^{(b)}_{t,j} - \mu^{(b)}_j}{\sigma^{(b)}_j + \varepsilon}, \quad \varepsilon = 10^{-8}. \tag{3}$$

Targets are winsorized and clipped as in Section 2.

3.2 | Fourier Encoding With Energy Truncation

For each branch, we apply a discrete Fourier transform (DFT) along time. With $\mathrm{i}^2 = -1$ and $F_b = \lfloor T_b/2 \rfloor + 1$,

$$\hat{X}^{(b)}_{f,j} = \sum_{t=0}^{T_b-1} \tilde{X}^{(b)}_{t,j}\, e^{-\,\mathrm{i}\,2\pi f t/T_b}, \qquad f = 0, \ldots, F_b - 1,\ j = 1, \ldots, d_b. \tag{4}$$

We concatenate the real and imaginary parts and apply a Hann smoothing (window $w = 3$) to magnitudes. We keep the smallest K_b whose cumulative spectral energy reaches τ (default $\tau = 0.95$), with caps $K^{\max}_{\text{year}} = 8$, $K^{\max}_{\text{quarter}} = 12$, $K^{\max}_{\text{month}} = 16$:

$$K_b = \min\left\{ k : \frac{\sum_{f=0}^{k-1} \sum_j |\hat{X}^{(b)}_{f,j}|^2}{\sum_{f=0}^{F_b-1} \sum_j |\hat{X}^{(b)}_{f,j}|^2} \geq \tau \right\}. \tag{5}$$

Stacking retained frequencies yields $Z^{(b)} \in \mathbb{R}^{K_b \times (2d_b)}$.

3.3 | Branch Encoders and Fusion

Each branch passes a lightweight 1D-CNN (kernel size 3, channels $c \in \{32, 64\}$, GELU), followed by global average pooling:

$$h_b = \mathrm{GAP}\big(\mathrm{Conv1D}_L(Z^{(b)})\big) \in \mathbb{R}^c. \tag{6}$$

Let $H = [h_{\text{year}}; h_{\text{quarter}}; h_{\text{month}}] \in \mathbb{R}^{3c}$. A softmax gate produces normalized weights and the fused representation:

$$\alpha = \mathrm{softmax}(WH + b) \in \mathbb{R}^3, \qquad h = \sum_b \alpha_b\, h_b \in \mathbb{R}^c. \tag{7}$$

Prediction head:

$$\hat{y}_{t+1} = w^\top h + b_0. \tag{8}$$

3.4 | Objective and Regularization

We minimize MSE with ℓ_2 weight decay using Adam (10^{-3}), weight decay 10^{-4}, batch size 16, dropout 0.2:

$$\mathscr{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2 + \lambda \|\theta\|_2^2, \quad \lambda = 10^{-4}. \tag{9}$$

Early stopping monitors validation loss (patience 8). Each experiment uses three seeds; we report mean ± SD.

4 | Evaluation Protocol

4.1 | Rolling-Origin Expanding-Window

At each origin $t \in \mathscr{T}$, models train on $[1,\ t - v]$, validate on $(t - v,\ t]$ (length v, default $v = 3$), and forecast $t + 1$ one step ahead. Preprocessing (standardization, imputation, Fourier encoding) is refit on the training window *at each origin* to avoid look-ahead leakage. The rolling-origin expanding-window protocol is illustrated in Figure 1.

4.2 | Temporal Alignment and Leakage Control

For each branch $b \in \{\text{year}, \text{quarter}, \text{month}\}$, only observations with timestamps $\leq t$ are visible at origin t. Quarterly/monthly streams are encoded along their own time axes and then aligned to the annual forecast time; no future filings beyond t are used. Targets are winsorized/clipped as in Section 2; the same processed target is used for training and evaluation.

4.3 | Per-Origin Training and Early Stopping

Within each origin, optimization uses Adam (10^{-3}), weight decay 10^{-4}, batch size 16, dropout 0.2, with early stopping on

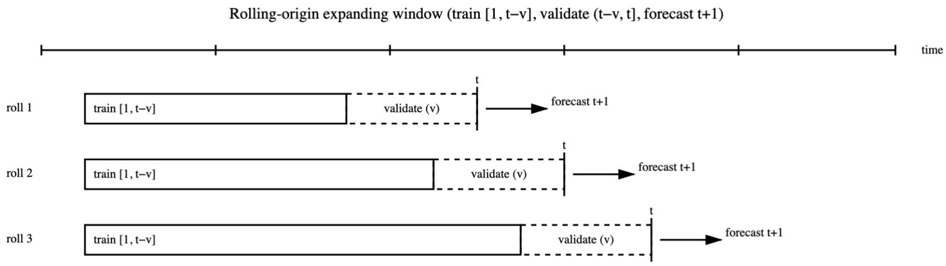

FIGURE 1 | Rolling-origin expanding-window. At each origin t, models train on $[1,\ t-v]$, validate on $(t-v,\ t]$, and forecast $t+1$. Metrics are recorded at every roll and aggregated across origins. The protocol is identical for all methods.

TABLE 1 | Quantitative comparison across baselines and our TiF model.

Category	Model	$R^2\uparrow$	RMSE↓	MAPE↓
Traditional	Linear regression Kogan et al. (2009)	0.9024	1.3182	6.3317
	Ridge regression Hoerl and Kennard (1970)	0.9831	1.3253	**1.4181**
	Decision tree Quinlan (1986)	0.9525	1.8691	6.4679
	Random forest Breiman (2001)	0.9338	2.8614	2.5347
	Gradient boosting Friedman (2001)	0.8718	2.4438	1.5855
	XGBoost Chen and Guestrin (2016)	0.8718	1.8105	9.5400
	LightGBM Ke et al. (2017)	0.8581	3.2533	9.6907
	CatBoost Dorogush et al. (2018)	0.9713	**1.1277**	8.2756
Deep	MLP Rosenblatt (1958)	0.9342	1.8147	3.7415
	LSTM Hochreiter and Schmidhuber (1997)	0.9491	2.1486	1.8790
	GRU Cho et al. (2014)	0.8529	2.5523	7.1581
	Transformer Vaswani et al. (2017)	**0.9858**	4.0333	4.9614
Mixed	Stacking Wolpert (1992)	0.9665	1.3985	2.0983
	Bagging Breiman (1996)	0.8797	2.8141	5.4566
Ours	TiF (Ours)	0.9585	4.5919	3.3290

Note: Best results per metric are **bolded**. ↑ indicates higher is better; ↓ indicates lower is better.

validation loss (patience 8). Each experiment is repeated with three seeds; we aggregate across origins and seeds.

4.4 | Metrics

For method m, seed s, and origin t, let $\hat{y}_{t+1}^{(m,s)}$ be the forecast and y_{t+1} the target (after preprocessing). Define $e_t^{(m,s)} = y_{t+1} - \hat{y}_{t+1}^{(m,s)}$. We report:

$$\mathrm{R}^2(m) = 1 - \frac{\sum_{t\in\mathcal{T}} \sum_s \left(e_t^{(m,s)}\right)^2}{\sum_{t\in\mathcal{T}} \left(y_{t+1} - \bar{y}\right)^2}, \quad \bar{y} = \frac{1}{|\mathcal{T}|} \sum_{t\in\mathcal{T}} y_{t+1}, \tag{10}$$

$$\mathrm{RMSE}(m) = \sqrt{\frac{1}{|\mathcal{T}|\ S} \sum_{t\in\mathcal{T}} \sum_s \left(e_t^{(m,s)}\right)^2}, \tag{11}$$

$$\mathrm{MAPE}(m) = \frac{100}{|\mathcal{T}|\ S} \sum_{t\in\mathcal{T}} \sum_s \frac{|e_t^{(m,s)}|}{|y_{t+1}| + \varepsilon}, \quad \varepsilon = 10^{-6}. \tag{12}$$

Here, $S = 3$ is the number of seeds. For cohort studies, metrics are computed per company and then summarized across companies as median (IQR) and mean $\pm$ SD.

4.5 | Hyperparameter Budgets and Selection

All baselines use the same rolling protocol and comparable tuning budgets (small grids per method). The best configuration per method is selected by validation RMSE averaged over origins; the selected configuration is then evaluated on the rolling test steps.

4.6 | Significance Testing

To compare TiF against the best non-TiF baseline, we run a paired, two-sided Wilcoxon signed-rank test on per-origin absolute errors. Let $\Delta_t = |e_t^{(\mathrm{TiF})}| - |e_t^{(\mathrm{best})}|$. The hypotheses are

$$H_0\colon\ \mathrm{median}(\Delta_t) = 0 \quad \text{vs.} \quad H_1\colon\ \mathrm{median}(\Delta_t) \neq 0. \tag{13}$$

We report p-values and mark tables with $*$ when $p < 0.05$. For multiple cohorts, Holm correction is applied across the corresponding tests.

4.7 | Reporting

We report mean $\pm$ SD across origins and seeds; additional diagnostics and company-wise summaries are provided in the appendices.

5 | Results and Discussion

We report results under the unified rolling-origin expanding-window protocol (Section 4), followed by external validations, ablations, and practical takeaways.

5.1 | Core Technology Benchmark (AAPL, MSFT, ADBE)

Annual targets with multi-timescale inputs (annual/quarterly/monthly) are processed exactly as in Section 3. Metrics (R^2, RMSE, MAPE) are aggregated across origins and seeds and reported per company and pooled.

5.1.1 | Quantitative Summary

A consolidated comparison across baselines and TiF is reported in Table 1. In addition, Figure 2 visualizes the per-metric performance (R2, RMSE, MAPE) together with a heatmap overview under the same rolling-origin protocol, matched preprocessing, and comparable tuning budgets.

5.1.2 | Observation

Across the three firms, TiF is consistently competitive and typically ranks at or near the top across metrics, offering a balanced trade-off rather than optimizing a single metric. Significance against the best non-TiF baseline is assessed via paired two-sided Wilcoxon tests (Section 4); stars in tables/figures indicate $p < 0.05$ when applicable.

5.2 | External Validation I: Non-Financial Cohort (EBITDA)

We apply the same protocol to the non-financial cohort in Section 2. Company-wise metrics are computed independently and summarized across firms (median/IQR and mean ± SD). Per-firm diagnostics are summarized in the appendices.

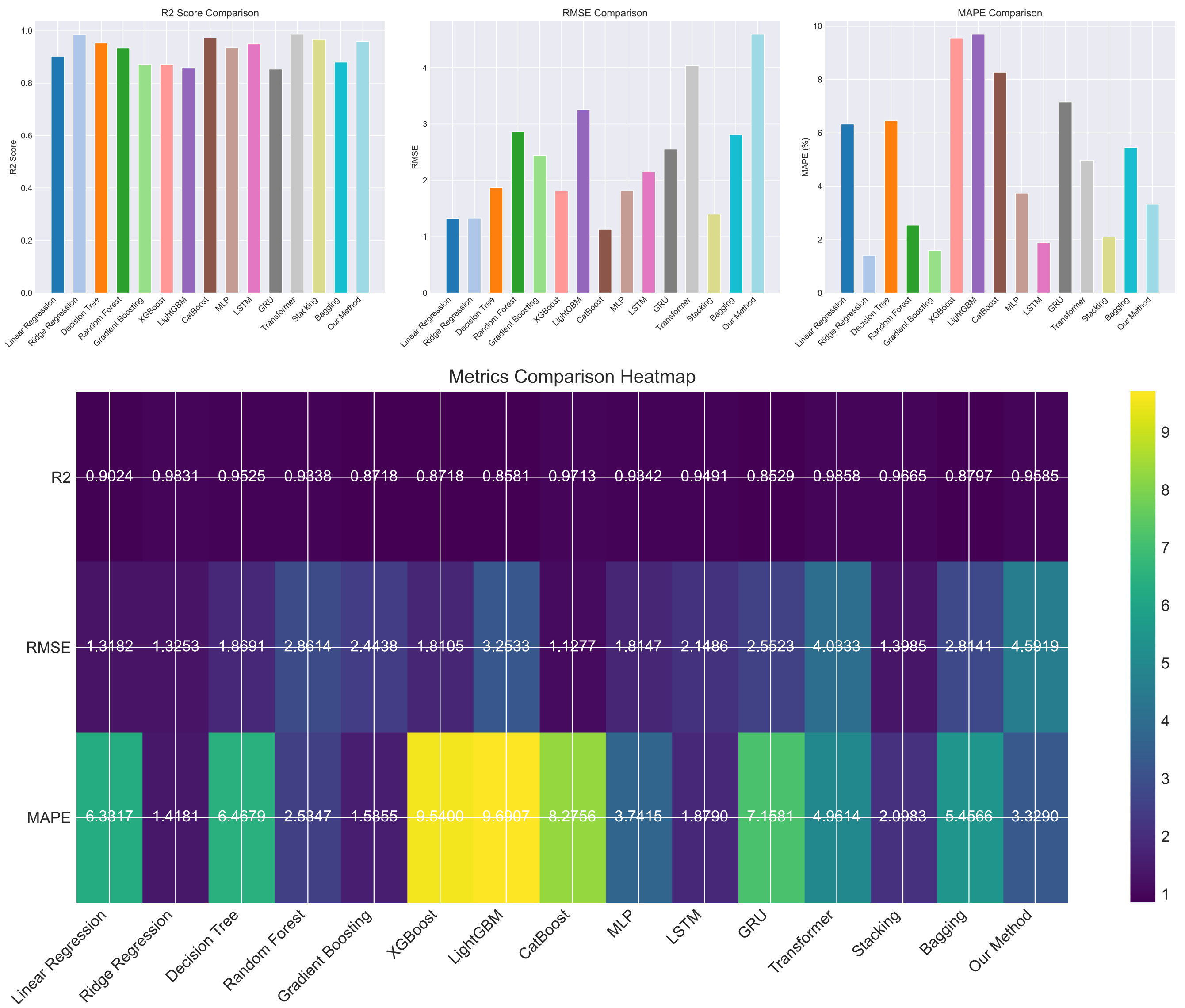

FIGURE 2 | Core benchmark (AAPL, MSFT, ADBE): comparison across baselines and TiF under the rolling-origin protocol. Top: bar charts (R2, RMSE, MAPE). Bottom: heatmap summary. All methods share identical preprocessing, validation length, and early-stopping/tuning budgets (Section 4).

5.3 | External Validation II: Financials (ROAA)

For finance-sector firms (ROAA target), the same screening and leakage controls apply. Stronger clipping and near-constant validation segments make this cohort more challenging; aggregate summaries remain robust, and representative firms with clearer variance structure show the same pattern as in the non-financial set. Company-level summaries are provided in the appendices; paired tests are reported whenever both TiF and the baseline have valid per-origin traces.

5.4 | Ablation and Sensitivity

We consider four ablative variants: removing the monthly branch (`NoMonthly`), removing the quarterly branch (`NoQuarterly`), removing annual PCA features (`NoPCA`), and using only PCA (`OnlyPCA`); all share the same protocol and budgets. The ablation results are summarized in Figure 3.

5.5 | Discussion and Practical Takeaways

5.5.1 | Balanced Accuracy Versus Simplicity

Compared with baselines that may peak on a single metric, TiF's multi-timescale fusion yields stable performance across R_2/RMSE/MAPE under matched budgets.

5.5.2 | Leakage-Safe Implementation

All preprocessing is refit per origin, which is crucial where reporting lags and calendar alignment matter.

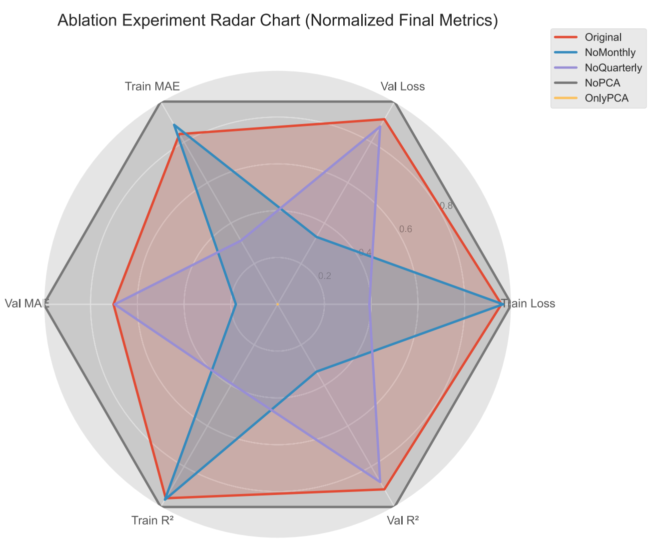

FIGURE 3 | Ablation radar chart (normalized metrics). Removing the monthly branch, removing the quarterly branch, removing annual PCA features, or using only PCA degrades at least one of R^2, RMSE, and MAPE, indicating complementary roles of multi-timescale inputs and the frequency-domain representation.

5.5.3 | When to Expect Gains

Firms exhibiting both low-frequency structure and higher-frequency variation benefit most from TiF's joint time/frequency design.

5.5.4 | Limits

Cohorts with strong clipping or near-constant validation segments compress observable differences; in such cases, we emphasize per-firm diagnostics and paired tests rather than global aggregates.

5.5.4.1 | Company-Level Summaries. Company-level median/IQR summaries for the external cohorts are provided in Appendix B: non-financial (EBITDA) in Table B1 and financials (ROAA) in Table B2. For each company, we select the epoch with the lowest validation RMSE and then aggregate R^2 and RMSE across companies.

6 | Reproducibility and Availability

All experiments share identical preprocessing, validation length, tuning budgets, and early-stopping criteria. We release (i) cohort CSVs with identifiers and screening metrics; (ii) `target_diagnose.csv`summarizing `val_std/clip_frac`; (iii) configuration files specifying winsor/clipping bounds and `min_years`; and (iv) scripts to rebuild cohorts from raw WRDS pulls. **Data access and licensing.** All Compustat/WRDS data are used under institutional license. **Code.** The full code is available at https://github.com/PUYUZHOU/TiF .

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used in this study are obtained from WRDS/Compustat under an institutional license and are therefore not publicly redistributable. To support reproducibility, the code and the scripts needed to reconstruct the cohorts from licensed WRDS pulls are available at https://github.com/PUYUZHOU/TiF.

Endnotes

[1] All data are used under institutional license; identifiers (e.g., `gvkey`, `conm`) are retained for reproducibility.

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Appendix A: Visualization of Target Variable

To illustrate the relationship between fundamentals and market performance, we visualize annual EBITDA alongside stock prices for three representative companies, as shown in Figure A1.

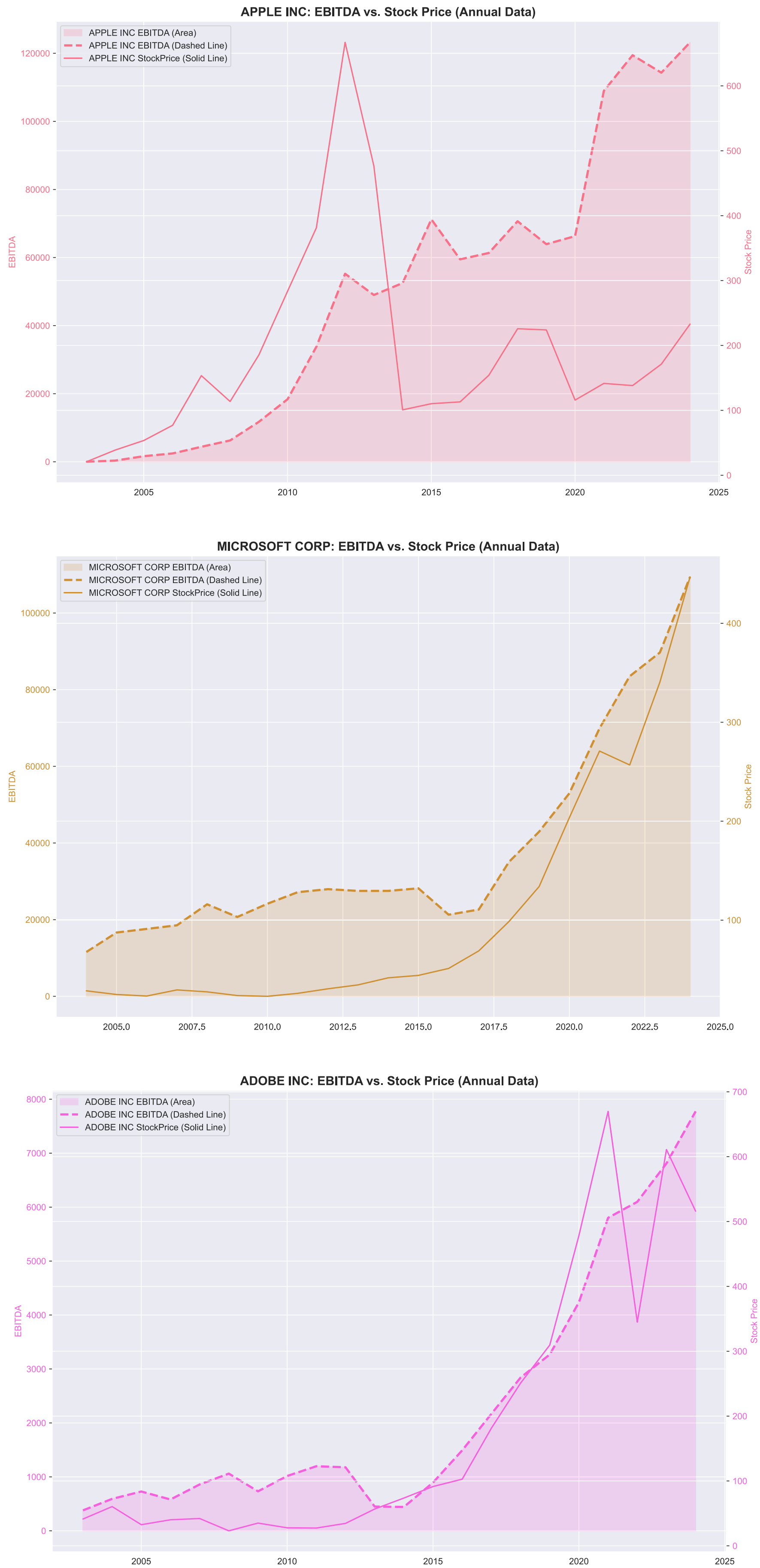

FIGURE A1 | Annual EBITDA and stock prices for (Top) Apple, (Middle) Microsoft, and (Bottom) Adobe.

Appendix B: Company-Level Summaries (Median/IQR)

We report company-wise metrics under the unified rolling-origin protocol and summarize across firms using median (IQR) and mean ± SD. For each company, we select the epoch with the lowest validation RMSE and then aggregate R^2 and RMSE over rolling steps.

TABLE B1 | Non-financial cohort (EBITDA): Company-level summary ($N = 3$).

Statistic	R^2	RMSE	% with $R^2 > 0$
Median (IQR)	−2.096 (−6.094–1.567)	0.089 (0.076–0.197)	0.0%
Mean ± SD	−4.409 ± 4.951	0.153 ± 0.133	—

Note: Best epoch per company chosen by lowest validation RMSE.

TABLE B2 | Financial cohort (ROAA): Company-level summary ($N = 3$).

Statistic	R^2	RMSE	% with $R^2 > 0$
Median (IQR)	−12.172 (−47.337–5.903)	0.011 (0.007–0.027)	33.3%
Mean ± SD	−31.436 ± 44.676	0.019 ± 0.021	—

Note: Best epoch per company chosen by lowest validation RMSE.

Appendix C: Per-Company Diagnostics and Cohort Roster

For transparency, we provide per-firm screening diagnostics, including history length, the validation-window standard deviation (`val_std`), and the clipping fraction (`clip_frac`). The diagnostics and final cohort rosters will be publicly released in our GitHub repository during/after review. Firms with excessive clipping or near-zero validation variance are removed according to the thresholds described in the main text.

Appendix D: Training Curves for Ablation Study

We provide training/validation curves for the ablations used in the paper as shown in Figure D1. Metrics include MSE loss, MAE, and R^2; all curves are computed under the same rolling-origin, early-stopping, and preprocessing settings.

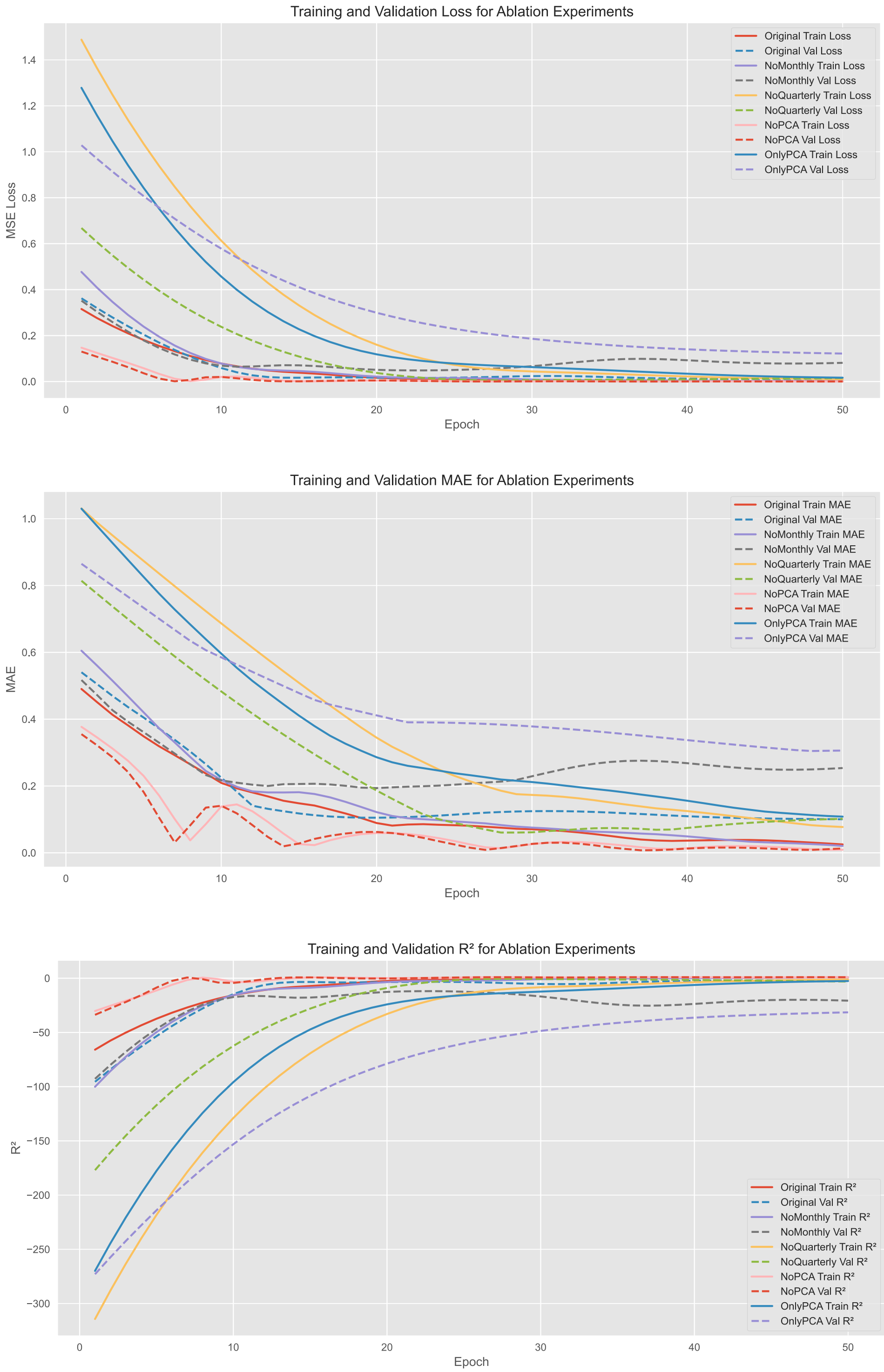

FIGURE D1 | Ablation training/validation curves. From top: MSE loss, MAE, and R^2.

Appendix E: Network Architecture and Hyperparameters

The TiF encoder uses a compact 1D-CNN with GELU activations, consistent with Section 3. For each time-scale branch (year/quarter/month), Fourier-encoded features with the real and imaginary parts concatenated are fed to a lightweight Conv1D stack followed by global average pooling. A learnable softmax gate produces cross-timescale fusion weights.

E.1 | Reference Configuration (Used Unless Stated)

- **Fourier truncation:** energy threshold $\tau = 0.95$, caps $K_{\text{year}}^{\max} = 8$, $K_{\text{quarter}}^{\max} = 12$, $K_{\text{month}}^{\max} = 16$; Hann smoothing window size $w = 3$.
- **CNN:** $L = 2$ conv layers per branch, kernel size 3, channels $c \in \{32, 64\}$, activation GELU, followed by GAP and a linear head.

- **Optimization:** Adam, lr 10^{-3}, weight decay 10^{-4}, batch size 16, dropout 0.2, early stopping patience 8.
- **Capacity:** ≈ 0.2 M–0.5 M parameters depending on $(K_b^{\max}, c)$.

Appendix F: Scalability and Generalization of the Multi-Timescale Fusion

The fusion module is scale-agnostic. Additional temporal slices (e.g., weekly or semi-annual) can be integrated as tensors $X^{(k)} \in \mathbb{R}^{T_k \times d_k}$ aligned on the calendar axis, passed through the same encoder, and included in the softmax gate without architectural changes. Cross-sectional hierarchies (regions, business units) can be injected as feature channels with near-linear compute overhead.

Appendix G: Financial-Theory Context and Cross-Disciplinary Insights

Monthly EBITDA reflects short-run cash-flow volatility; quarterly values reflect disclosure practices; annual values tie to valuation. Multi-timescale modeling separates transitory versus structural signals, while Fourier encoding disentangles high-frequency macro shocks from low-frequency industry cycles.

Appendix H: Rationale Behind Company Selection

We originally considered the top-10 US tech firms, but the full span (50+ years × 10 firms × 3 cadences) exceeded current compute limits. We therefore use three representative firms—Apple (hardware seasonality), Microsoft (enterprise + cloud), and Adobe (subscription model)—to retain diversity of operational rhythms while ensuring reproducible experiments.